جامعة المجمعة
Majmaah University
كلية علوم الحاسب والمعلومات

Brief Course Specifications

MS Cyber Security and Digital Forensics

Department of Information Technology

College of Computer and Information Sciences

Majmaah University

Al Majmaah, Saudi Arabia

Version: 2025 Effective Date: October 28, 2025

LEVEL 1:

IT 601 – Cybersecurity Foundations

• Course Description

This course provides fundamental overview of cybersecurity .Topics covered include: Historical evolution of cybersecurity threat. Core principles: CIA triangle (Confidentiality, Integrity, Availability) - Threat models and risk assessment frameworks (e.g., STRIDE, NIST), risk analysis Practical application of security policie. Overview of types of attack and defense mechanisms to protect from attack, Security Principles, security layers, Attacker Techniques and Exploit Tools. Exploitation and Defense and Analysis Techniques, Firewalls & Intrusion Detection System. Hands-on simulations of basic attacks and defenses - Use of virtual labs for threat identification. Security mechanism for secure communication.

• Learning Outcomes

- CLO1 : Student familiar with Cyber Security Fundamentals, Threat , Attacks, Risk ,Vurnabilities.
- CLO2: Gain knowledge of Attacker Techniques, various exploit Tools.
- CLO3: Risk Management, Risk Identification, Risk Assessment, Risk Control Strategies, Security policie, the ethics, and social issues related to cybersecurity
- CLO4: Defence system , Intrusion Detection System.
- CLO5: Design, implement security solutions for cybersecurity

• Teaching Strategies

Lectures, Hands-on labs, simulations, hybrid sessions

• Assessment Methods

Quiz, Midterm exam, Final exam, Lab assignments, Mini Project, Case study, Research paper

IT 602 – Digital Forensics Technology & Practices

• Course Description

This course provides overview of digital forensic. Topics covered includes, introduction to digital forensic, digital forensic process, digital forensic tools, investigative methodology, current techniques and tools for forensic examinations & Analysis, Digital investigations, Electronic Discovery ,Intrusion Investigation , Ant forensics, Windows Forensic Analysis ,Embedded Systems Analysis &Network Evidence and Investigations.

• Learning Outcomes

- Gain knowledge about roles, functions and services of operating systems, digital forensics, forensic process, anti-forensics, e-discovery, and forensic tools.
- Understand and apply investigative methodology and current techniques/tools for forensic examinations.
- Implement and analyze Windows/Linux forensic analysis.
- Analyze and apply forensic principles for collecting/detecting network evidence and design corrective procedures.
- Analyze business, legal, ethical, and case-specific requirements for forensic investigations.

• Teaching Strategies

Classroom delivery, guided discussion, lectures + lab demonstrations, hands-on labs, simulation, case studies

• Assessment Methods

Quiz, Assignment, Midterm, Final Exam, Lab Assignments, Lab Exam, Project Report, Presentations

IT 603 – Information Security Law, Policy, and Ethics

• Course Description

This course provides in depth analysis of information security laws in many jurisdictions and needs to understand the overall framework of legal security requirements along with Cyber Risk Assessment and Analysis Methodologies, it can evaluate how local law fits in, and what it might do to become generally legally compliant in many jurisdictions and under many laws. This course also provides a concise overview of the traditional ethical frameworks that can guide our analysis of the moral dilemmas and social problems that rise in cyberspace. Discussions on Saudi securities laws are also appended.

• Learning Outcomes

- Understand the need of information security law and ethics.
- Understand and analyze information security law & policy and how risk relates to security policy.
- Demonstrate the main risk analysis methodologies.
- Analyze and compare various information security law standards.
- Choose appropriate methodology for dealing with cyber risks.

• Teaching Strategies

Classroom lectures, case study discussion, applied case studies, group discussions, simulations

• Assessment Methods

Quizzes, Mid-term, Final Exam, Case study report, Group Project, Presentation, Reflection paper

LEVEL 2:

IT 604 – Cybersecurity Technology and Management

• Course Description

This course provides issues in cyber security technology and practice. This course explores practical solutions to a wide range of cyber-physical and IT infrastructure protection issues. This course also aims at managing the cyber-attacks on the critical IT infrastructure computer networks.

• Learning Outcomes

- Understand and analyze the issues in Cyber Security Technology
- Study and analyze the practical solutions and IT infrastructure protection issues
- Managing the Cyber Attack Search and Destroy
- Identify the breached design principle associated with common system security weaknesses
- Collaborative project development

• Teaching Strategies

Classroom Teaching, Lab based teaching

• Assessment Methods

Quizzes, Midterm Exam, Final Exam, Lab based assignment, Mini Project, Seminar Presentation

IT 612 – Malware and Vulnerability Analysis

• Course Description

This course aims to to identify, analyse, and exploit software vulnerabilities, focusing on application-level issues which includes Code auditing, reversing, memory corruption, fuzzing, Post-Exploitation, Web app Hacking, Reversing, Reverse Engineering, Client-side attacks and Web Hacking on android and smart devices.

• Learning Outcomes

- Understand the Application Structure and security aspects
- Understand Static & dynamic approaches

- Understand Reverse Engineering in Android
- Understand Anti-Reverse Engineering in Android
- Analyse mobile applications using reverse engineering tools

• Teaching Strategies

Classroom delivery, guided discussion, Hands-on labs, simulation

• Assessment Methods

Quiz, Assignment, Midterm, Final Exam, Lab Assignments, Lab Exam

IT 614 – Software Security

• Course Description

This course will provide students to understand the theories and tools used for secure software design, threat analysis, secure coding, and vulnerability analysis.

• Learning Outcomes

- Describe the risks, threats, and vulnerabilities
- Understand how to protect software
- Address issues in Web applications Security
- Apply code auditing practices
- Integrate technologies in software security

• Teaching Strategies

Classroom delivery, guided discussion, labs, simulation, case studies

• Assessment Methods

Quiz, Assignment, Midterm, Final Exam, Lab Assignments, Project Report

IT 616 – Special Topics in Cybersecurity

• Course Description

This course provides an introduction to the fundamental concepts, principles, and technologies behind blockchain systems.

• Learning Outcomes

- Understand blockchain concepts

- Understand cryptographic foundations
- Achieve proficiency with blockchain apps
- Analyze blockchain threats

• Teaching Strategies

Lectures, lab demonstrations, hands-on labs, simulations

• Assessment Methods

Quiz, Assignment, Lab Exam, Final Exam, Lab Assignments, Project Report

IT 623 – Mobile Network and Forensics

• Course Description

This course aims to provide mobile and network forensic investigations that are legal, ethical, and highly effective.

• Learning Outcomes

- Understand Mobile and Network Forensics
- Prepare and protect digital evidence
- Understand intrusion detection systems
- Describe network logs and artifacts
- Analyze mobile and network data

• Teaching Strategies

Classroom delivery, guided discussion, hands-on labs, simulation

• Assessment Methods

Quiz, Assignment, Midterm, Final Exam, Lab Assignments, Lab Exam

IT 624 – Malware Development and Countermeasures

• Course Description

This course aims to provide malware development strategies and defence strategies such as threat modelling, static and dynamic application security testing.

• Learning Outcomes

- Understand malware analysis techniques

- Apply anti-analysis techniques
- Use tools like IDA Pro, OllyDbg
- Analyze host and network indicators
- Understand executable formats and APIs
- Ethical practices in malware analysis

• Teaching Strategies

Classroom delivery, guided discussion, hands-on labs, simulation

• Assessment Methods

Quiz, Assignment, Midterm, Final Exam, Lab Assignments, Lab Exam, Project Report

IT 625 – Forensics Management of Digital Evidence

• Course Description

This course provides an in-depth understanding of managing digital evidence through its lifecycle from acquisition to disposal.

• Learning Outcomes

- Understand evidence lifecycle models
- Implement quality standards
- Manage evidence tracking systems
- Conduct forensic risk assessments
- Perform audits and compliance

• Teaching Strategies

Lectures, lab demonstrations, hands-on labs, simulation

• Assessment Methods

Quiz, Assignment, Midterm, Final Exam, Lab Assignments, Lab Exam

LEVEL 3:

IT 605 – Research Methods and Engineering Statistics

• Course Description

Study of the research methods and processes applicable to engineering/technology Emphasis on defining research problems, collecting, analyzing, recording, and interpreting data. Students will be required to conduct a research project.

• Learning Outcomes

- Explain the fundamental research paradigms (qualitative vs. quantitative).
- Understand core statistical concepts (hypothesis testing, regression, sampling).
- Apply statistical software tools to analyze cybersecurity datasets.
- Design a research experiment and test hypotheses.
- Critically evaluate research findings and present project proposal.

• Teaching Strategies

Lectures, problem-solving demonstrations, workshops, project-based learning, seminars

• Assessment Methods

Quizzes, Assignments, Project reports, Demonstrations, Presentations

IT 616 – Special Topics in Cybersecurity

• Course Description

This course provides an introduction to the fundamental concepts, principles, and technologies behind blockchain systems. Students will explore the decentralized nature of blockchains, their applications beyond cryptocurrencies, and the underlying cryptographic techniques used for securing transactions.

• Learning Outcomes

- Understand core blockchain concepts.
- Understand cryptographic foundations and consensus mechanisms.
- Achieve proficiency with blockchain networks and applications.
- Analyze blockchain and investigate threats.

• Teaching Strategies

Lectures, lab demonstrations, hands-on labs, simulations, case studies

• Assessment Methods

Quiz, Assignment, Lab Exam, Final Exam, Lab Assignments, Project Report

IT 621 – Computer Forensics and Investigations

• Course Description

This course gives students a solid foundation to the methods of computer forensics and investigations. It provides an in-depth knowledge of the criminal justice system, Computer hardware and software systems, investigative and evidence gathering protocols.

• Learning Outcomes

- Demonstrate knowledge of computer forensic principles.
- Identify forensic tools and methodologies.
- Apply forensic tools to analyze digital evidence.
- Develop forensic investigation reports.
- Evaluate legal and ethical implications.

• Teaching Strategies

Lectures, case-based discussions, labs, simulations, report-writing workshops

• Assessment Methods

Midterm exam, final exam, quizzes, lab reports, group project, presentations

LEVEL 4:

IT 650 – Research Project

• Course Description

Students will conduct research under supervision and submit a graduate research project or dissertation.

• Learning Outcomes

- Understand security and forensics concepts
- Conduct literature analysis
- Design and implement cybersecurity solutions
- Utilize resources and collaboration
- Present research outcomes

• Teaching Strategies

Supervised research, independent study, project-based learning

• Assessment Methods

Research report, project evaluation, presentation, supervisor assessment